Econ 590: News Driven Business Cycles (II)

Paul Beaudry & Franck Portier

University of British Columbia & Toulouse School of Economics

March 24, 2016

0. Introduction

- Build on section 4 of *"News Driven Business Cycles: Insights and Challenges*", Journal of Economic Literature, 2014.

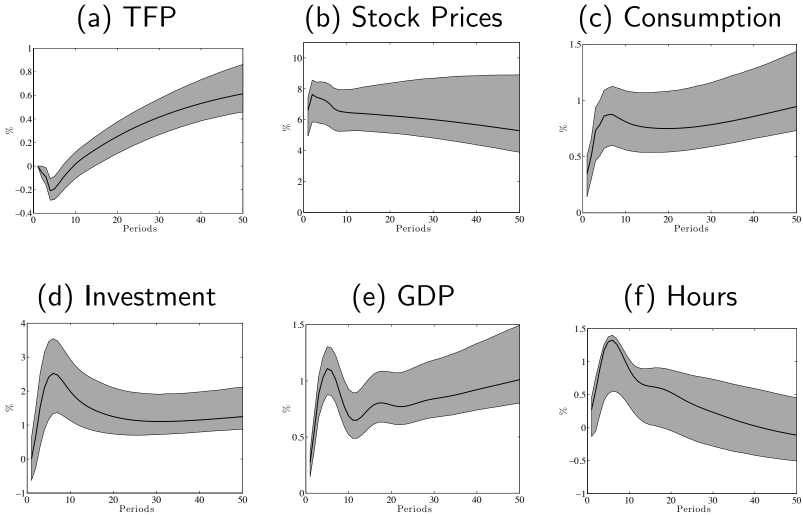
0. Introduction

Challenge facing Perception driven business cycles (News being a special case)

- Defining a perception driven business cycle as one that does not change current fundamentals, but affects our perception of the future.
- We will emphasize two challenges
- A Dynamic challenge, getting good news to favor increasing investment now
- A static challenge. Have the increased desire to invest also increase current consumption.
- We want to have good news increase both consumption and investment as this is what we believe is supported by the data (mention max-var results).

Recall: Identification in VARs

In two slides: response to a TFP news



- Simple U-V set in one sector model

$$\max_{C_t, K_{t+1}, L_t} U(C_t, L_t) + \beta E\left[\widetilde{V}(K_{t+1}, \Gamma_{t+1}) \mid \Omega_t\right],$$
$$\text{s.t.} \qquad C_t + K_{t+1} = w_t L_t + K_t(1 - \delta + r_t) + \pi_t,$$

- Define

$$V(K_{t+1}, \theta_t, S_t) = \beta E\left[\widetilde{V}(K_{t+1}, \Gamma_{t+1}) \mid \Omega_t = \{\theta_t, S_t\}\right].$$

- Assume

$$\frac{\partial^2 V(K_{t+1}, \theta_t, S_t)}{\partial K_{t+1} \partial S_t} > 0.$$

0. Introduction

- F.O.C.

$$
\begin{aligned}
\frac{\partial V(K_{t+1}, \theta_{t+1}, S_t)]}{\partial K_{t+1}} &= U_C(C_t, L_t), && (1)\\
F_L(K_t, L_t, \theta_t) &= \frac{-U_L(C_t, L_t)}{U_C(C_t, L_t)}, && (2)\\
C_t + K_{t+1} &= F(K_t, L_t, \theta_t) + (1-\delta)K_t. && (3)
\end{aligned}
$$

- Result: increase in S_t can lead to increase in both C_t and I_t as long as preferences for consumption and leisure are normal, and production function satisfies CRS. This is the static problem.

0. Static Challenge

- The sticky price solution

$$\max_{C_t, K_{t+1}, M_{t+1}, L_t} U(C_t, L_t) + \beta E[V(K_{t+1}, M_{t+1}, \theta_t, S_t),$$

$$\text{s.t.} \quad C_t + K_{t+1} + \frac{M_{t+1}}{P_t} = w_t L_t + (1 - \delta + r_t)K_t + \frac{M_t}{P_t} + \pi_t + \frac{\tau_t}{P_t},$$

- where τ_t is an exogenous money transfer, with $M_{t+1} = M_t + \tau_t$.
- suppose policy rule of form

$$\tau_t = \alpha(Y_t - \overline{Y})$$

0. Static Challenge

- Equilibrium conditions for C_t, I_{t,L_t} (assuming fixed prices)

$$
\begin{aligned}
V_K(K_{t+1}, M_{t+1}, \theta_t, S_t) &= V_M(K_{t+1}, M_{t+1}, \theta_t, S_t), \\
U_C(C_t, L_t) &= V_K(K_{t+1}, M_{t+1}, S_t), \\
F(K_t, L_t) &= C_t + K_{t+1} - (1-\delta)K_t, \\
M_{t+1} &= M_t + \alpha(C_t + I_t - \overline{Y}), \\
I_t &= K_{t+1} - (1-\delta)K_t.
\end{aligned}
$$

- : increase in S_t leads to generalized boom only if $\alpha > 0$ (easy to show with separability)
- drawbacks: effects depend entirely on policy (and often sub-optimal policy) and should cause inflation

0. Static Challenge

- Two sector model (CRS)
- $C_t = F(K_t^C, L_t^C; \theta_t^C)$ and $I_t^I + I_t^C = G(K_t^I, L_t^I; \theta_t^I)$,
- Household problem

$$\max_{C_t, K_{t+1}^C, K_{t+1}^I, L_t^C, L_t^I} U(C_t, L_t^C, L_t^I) + V(K_{t+1}^C, K_{t+1}^I, \theta_{t+1}^C, \theta_{t+1}^I, S_t),$$

$$\text{s.t.} \quad C_t + P_t^I(K_{t+1}^C + K_{t+1}^I) =$$

$$w_t^C L_t^C + w_t^I L_t^I + K_t^C(P_t^I(1-\delta) + r_t^C) + K_t^C(P_t^I(1-\delta) + r_t^I)$$

$$+ \pi_t^C + \pi_t^I,$$

0. Static Challenge

- FOC consumption sector

$$
\begin{aligned}
\frac{-U_{L^C}(L_t^C)}{U_C(C_t)} &= F_{L^C}(K_t^C, L_t^C), \\
C_t &= F(K_t^C, L_t^C).
\end{aligned}
$$

- FOC Investment sector

$$
\begin{aligned}
G_{L^I} &= \frac{-U_{L^I}}{V_{K^I}}, \\
G_{L^I} &= \frac{-U_{L^I}}{V_{K^C}}, \\
G(K_t^I, L_t^I) &= K_{t+1}^I - (1-\delta)K_t^I + K_{t+1}^C - (1-\delta)K_t^C.
\end{aligned}
$$

- can easily get investment to increase, without fall in consumption. But no increase. If economies of scope are introduced, then possible to get both.

- Adding L_t in value function: adjustment costs.

$$\max_{C_t, K_{t+1}, L_t} U(C_t, L_t) + V(K_{t+1}, L_t, \theta_t, S_t),$$

$$\text{s.t.} \qquad C_t + K_{t+1} = w_t L_t + K_t(1 - \delta + r_t) + \pi_t,$$

0. Static Challenge

- Labor market condition

$$-U_C(C_t, L_t)F_L(K_t, L_t, \theta_t) = U_L(C_t, L_t) + V_L(K_{t+1}, L_t, \theta_t, S_t)$$

- Fully differentiating and assuming separability of utility

$$\underbrace{(U_C F_{LL} + U_{LL} + V_{LL})}_{-} dL_t = \underbrace{-U_{CC}}_{+} dC_t + \underbrace{-V_{LK}}_{-} dK_{t+1} + \underbrace{-V_{LS}}_{-} dS_t$$

- so possible to get both C_t and L_t to rise.
- Drawback: mechanism weak and not likely very persistent.

Summary: Static Challenge

- To get perception to lead to generalized and sustained boom, without causing inflation, need to depart from standard framework.
- We will later discuss: moving away from representative agent, CRS and Walrasian setup.
- Look at dynamic challenge

Dynamic Challenge

- Focus on two sector case (one sector is special case)
- Value function defined by

$$
\begin{aligned}
\widetilde{V}(K_t^C, K_t^I, \Sigma_t) = & \max_{\{C_{t+i}, K_{t+1+i}^C, K_{t+1+i}^I, L_{t+i}^C, L_{t+1+i}^I\}_{i=0}^{\infty}} E_t \sum_{i=0}^{\infty} U(C_{t+i}, L_{t+i}^C, L_{t+i}^I), \\
\text{s.t.} \quad & C_{t+i} = F(K_{t+i}^C, L_{t+i}^C, \theta_{t+i}^C), \\
& (K_{t+1+i}^C - (1-\delta)K_{t+i}^C) + (K_{t+1+i}^I - (1-\delta)K_{t+i}^I) = G(K_{t+i}^I, L_{t+i}^I, \theta_{t+i}^I),
\end{aligned}
$$

- where $\Sigma_t = (\theta_t^C, \theta_t^I, \epsilon_t^C, \epsilon_t^I)$
- envelope condition

$$
\begin{aligned}
\widetilde{V}_{K^C}(K_{t+1}^C, K_{t+1}^I, \Sigma_{t+1}) = & U_C(C_{t+1}, L_{t+1}, L_{t+1}^I) F_{K^C}(K_{t+1}^C, L_{t+1}^C, \theta_{t+1}^C) \\
& + \frac{(1-\delta)}{G_{L^I}(K_{t+1}^I, L_{t+1}^I, \theta_{t+1}^I)} - U_{L^I}(C_{t+1}, L_{t+1}^C, L_{t+1}^I)
\end{aligned}
$$

and

$$
\begin{aligned}
\widetilde{V}_{K^I}(K_{t+1}^C, K_{t+1}^I, \Sigma_{t+1}) = & \frac{-U_{L^I}(C_{t+1}, L_{t+1}, L_{t+1}^I)}{G_{L^I}(K_{t+1}^I, L_{t+1}^I, \theta_{t+1}^I)}((1-\delta) \\
& + G_{K^I}(K_{t+1}^I, L_{t+1}^I, \rho\theta_{t+1}^I)).
\end{aligned}
$$

$$
\begin{aligned}
\widetilde{V}_{K^I}(K_{t+1}^C, K_{t+1}^I, \Sigma_{t+1}) = & \frac{-U_{L^I}(C_{t+1}, L_{t+1}, L_{t+1}^I)}{G_{L^I}(K_{t+1}^I, L_{t+1}^I, \theta_{t+1}^I)}((1-\delta) \\
& + G_{K^I}(K_{t+1}^I, L_{t+1}^I, \rho\theta_{t+1}^I)).
\end{aligned}
$$

Dynamic Challenge

- Assuming utility of form $\ln C_t - \nu \cdot (L_t^C + L_t^I)$, and cobb-douglas production functions with capital shares α^C and α^I

$$V_{K^C}(K_{t+1}^C, K_{t+1}^I, \Sigma_{t+1}) = \beta\left(\frac{\alpha^C}{K_{t+1}^C} + \frac{\nu(1-\delta)E_t(L_{t+1}^I)^{\alpha^I}}{(1-\alpha^C)(\theta_t^I + \epsilon_t^I)(K_{t+1}^I)^{\alpha^I}}\right), \quad (4)$$

$$V_{K^I}(K_{t+1}^C, K_{t+1}^I, \Sigma_{t+1}) = \beta\left(\frac{\nu(1-\delta)E_t(L_{t+1}^I)^{\alpha^I}}{(1-\alpha^C)(\theta_t^I + \epsilon_t^I)(K_{t+1}^I)^{\alpha^I}} + \frac{\nu E_t(L_{t+1}^I)}{K_{t+1}^I}\right). \quad (5)$$

- : Notice: depends only on endogenous variable $L_{I_{t+1}}$, ϵ^C does not appear, ϵ^I only appears in denominator.

Dynamic Challenge

- What to take away
- Difficult to get news increase the desire to invest with standard parameterization.
- Two potential avenues: (1) reduce sensitivity of discount factor to to expected growth (different agents type, or other preferences ex: Epstein-Zin), (2) Change production structure so that effect of technology affect marginal gains to investment more than average gains.

Dynamic Challenge: Alternative approach

- Jaimovich and Rebello (AER 2009) approach: flip everything on it head. Have news decrease desire to accumulate, but still get consumption and investment to increase.
- Advantage: very close to standard model.
- Main components: investment adjustments cost, capacity utilization, and much much less important, preference structure (put aside here).

Dynamic Challenge: JR

- Household problem

$$\max_{\{C_{t+i},K_{t+1+i},L_{t+i},\mu_{t+i}\}_{i=0}^{\infty}} E_t \sum_{i=0}^{\infty} U(C_{t+i}, L_{t+i}),$$

$$\text{s.t} \quad C_{t+i} + K_{t+1+i} = F(\mu_{t+i}K_{t+i}, L_{t+i}, \theta_{t+i}) + (1-\delta(\mu_{t+i}))K_{t+i} - \frac{\psi}{2}(I_{t+i} - I_{t-1-i})^2,$$

$$I_{t+i} = K_{t+1+i} - (1-\delta(\mu_{t+i}))K_{t+i},$$

- Can re-write

$$\max_{\{C_t,K_{t+1},I_t,L_t,\mu_t\}} U(C_t, L_t) + V(K_{t+1}, I_t, \Theta_t, \epsilon_t),$$

$$\text{s.t.} \quad C_t + K_{t+1} = F(\mu_t K_t, L_t, \theta_t) + (1-\delta(\mu_t))K_t - \frac{\psi}{2}(I_t - I_{t-1})^2,$$

$$I_t = K_{t+1} - (1-\delta(\mu_t))K_t.$$

Dynamic Challenge: JR

- equilibrium outcomes for $C_t, K_{t+1}, \mu_t, L_t, I_t$ is then given as the solution to the following equations plus the two previous constraints

$$F_\mu(\mu_t K_t, L_t, \theta_t) = K_t \delta_\mu(\mu_t) \frac{V_K(K_{t+1}, I_t, \Theta_t, \epsilon_t)}{U_C(C_t, Lt)},$$

$$\frac{V_I(K_{t+1}, I_t, \Theta_t, \epsilon_t)}{U_C(C_t, Lt)}) = \psi(I_t - I_{t-1}) + \left(1 - \frac{V_K(K_{t+1}, I_t, \Theta_t, \epsilon_t)}{U_C(C_t, Lt)}\right),$$

$$\frac{-U_L(C_t, L_t)}{U_C(C_t, Lt)} = F_L(\mu_t K_t, L_t, \theta_t).$$

- first equation is the determination of utilization
- Note: ϵ news decreases V_K, but increases incentive to invest. Optimal response: increase utilization. This increase MP of labor, so can get both I and C to increase.

Dynamic Challenge: JR

- Note: Story very different, even if reduce delivers.
- Drawbacks(?): Flipped implication for stock prices and price of capital.
- Noise does not lead to over-accumulation but to under-accumulation
- Look at multi-agent gains from trade approach.